Matrices

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1 Introduction to Matrices

The order of a matrix is the number of rows multiplied by the number of columns.

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \text{Order of } B = 2 \times 2.$$

1.1 Types of Matrices

1. Row matrix: order $1 \times n$.
2. Column matrix: order $m \times 1$.
3. Rectangular matrix: $m \neq n$.
4. Square matrix: $m = n$.
5. Diagonal matrix: $a_{ij} = 0$ when $i \neq j$; for example $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$.
6. Scalar matrix: all diagonal entries are equal, for example $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.
7. Identity or unit matrix: all diagonal entries are 1, for example $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
8. Null or zero matrix: every entry is zero, for example $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

2 Operations on Matrices

2.1 Addition and Negation

$$A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad A + B = \begin{bmatrix} 4 & 4 \\ 1 & 3 \end{bmatrix}.$$

$$\text{If } A = \begin{bmatrix} 3 & 4 \\ -1 & 2 \end{bmatrix}, \quad -A = \begin{bmatrix} -3 & -4 \\ 1 & -2 \end{bmatrix}.$$

2.2 Scalar Multiplication

$$k \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3k & 4k \\ k & 2k \end{bmatrix}, \qquad 2A = \begin{bmatrix} 6 & 8 \\ 2 & 4 \end{bmatrix}.$$

2.3 Matrix Multiplication

If A has order $m \times p$ and B has order $p \times n$, then AB is defined and has order $m \times n$. The number of columns of the first matrix must equal the number of rows of the second.

$$A_{m \times p} B_{p \times n} = (AB)_{m \times n}.$$

3 Transpose

The transpose interchanges rows and columns.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

3.1 Transpose Properties

$$(A^T)^T = A$$
$$(kA)^T = kA^T$$
$$(A+B)^T = A^T + B^T$$
$$(AB)^T = B^T A^T.$$

4 Symmetric and Skew Symmetric Matrices

A square matrix is symmetric if $A^T = A$ and skew symmetric if $A^T = -A$.

$$\begin{bmatrix} 1 & a & b \\ a & c & d \\ b & d & e \end{bmatrix} \text{ is symmetric,} \qquad \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} \text{ is skew symmetric.}$$

A skew symmetric matrix has zero diagonal entries.

4.1 Decomposition of a Square Matrix

For any square matrix A, let

$$P = \frac{A + A^T}{2}, \qquad Q = \frac{A - A^T}{2}.$$

Then P is symmetric, Q is skew symmetric, and

$$\boxed{A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = P + Q.}$$

5 Invertible Matrices

Square matrices A and B are inverses if $AB = BA = I$. Then $B = A^{-1}$.

$$(AB)^{-1} = B^{-1}A^{-1}.$$